REFERENCE FILE

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REPORT DOCUMENTATION PAGE

READ INSTRUCTIONS BEFORE COMPLETING FORM

1. REPORT NUMBER: TOP 4-2-504	2. GOVT ACCESSION NO.: AD A070340
3. RECIPIENT'S CATALOG NUMBER:	
4. TITLE *(and Subtitle)*: US ARMY TEST AND EVALUATION COMMAND TEST OPERATIONS PROCEDURE, "SAFETY TESTING OF ARTILLERY, MORTAR, AND RECOILLESS RIFLE AMMUNITION"	5. TYPE OF REPORT & PERIOD COVERED: FINAL
	6. PERFORMING ORG. REPORT NUMBER:
7. AUTHOR(s):	8. CONTRACT OR GRANT NUMBER(s):
9. PERFORMING ORGANIZATION NAME AND ADDRESS: US ARMY ABERDEEN PROVING GROUND (STEAP-MT-M) ABERDEEN PROVING GROUND, MARYLAND 21005	10. PROGRAM ELEMENT, PROJECT, TASK AREA & WORK UNIT NUMBERS: DARCOM-R 310-6
11. CONTROLLING OFFICE NAME AND ADDRESS: US ARMY TEST AND EVALUATION COMMAND (DRSTE-AD-M) ABERDEEN PROVING GROUND, MARYLAND 21005	12. REPORT DATE: 1 April 1979
	13. NUMBER OF PAGES: 39
14. MONITORING AGENCY NAME & ADDRESS*(if different from Controlling Office)*:	15. SECURITY CLASS. *(of this report)*: Unclassified
	15a. DECLASSIFICATION/DOWNGRADING SCHEDULE:

16. DISTRIBUTION STATEMENT *(of this Report)*

Approved for public release; distribution unlimited.

17. DISTRIBUTION STATEMENT *(of the abstract entered in Block 20, if different from Report)*

18. SUPPLEMENTARY NOTES

19. KEY WORDS *(Continue on reverse side if necessary and identify by block number)*

Ammunition, Artillery
Ammunition, Mortar
Ammunition, Recoilless Rifle
Safety

20. ABSTRACT *(Continue on reverse side if necessary and identify by block number)*

Describes safety evaluation test procedures applicable to all ammunition for field and air defense artillery, tank guns, recoilless rifles, and mortars. Although primarily oriented toward explosive-loaded projectiles, procedures for nonexplosive projectiles are included. Covers launch, flight, and nvironmental hazards as well as compatibility of the ammunition with the eapon system. Test phases include propellant checkout, metal parts checkout, torage test, transportation and rough handling tests, and supplemental tests. Excludes nuclear weapon projectiles.

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U. S. ARMY TEST AND EVALUATION COMMAND
Aberdeen Proving Ground, Maryland 21005

TOP 4-2-504
AD A070340
CHANGE 1

31 October 1979

SAFETY TESTING OF ARTILLERY, MORTAR, AND RECOILLESS RIFLE AMMUNITION

TOP 4-2-504, 1 April 1979, is changed as follows:

1. Remove pages and insert new pages as indicated below:

Remove pages--	Insert pages--
11 and 12	11 and 12

2. A vertical line in the margin indicates the changed portion of the revised page.

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US ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

DRSTE-RP-702-103
*Test Operations Procedure 4-2-504 1 April 1979
AD No. A070340

SAFETY TESTING OF ARTILLERY, MORTAR, AND RECOILLESS RIFLE AMMUNITION

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1. SCOPE. This TOP provides procedures for safety testing of developmental and production ammunition (except nuclear) for field and air defense artillery, tank guns, mortars, and recoilless rifles. These procedures are mandatory for validating design safety, and are normally repeated to establish safety of the production item during DT III, or first article testing. Procedures are applicable to complete rounds and components. Hazards related to ammunition handling, transport, launch, flight and environmental conditions, and the compatibility of the

*This TOP supersedes MTP 4-2-504, 11 December 1972.

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ammunition with the weapon system in terms of weapon safety are considered. Additional background information is contained in Appendix E. Test organization is shown in Appendix B.

2. FACILITIES AND INSTRUMENTATION.

2.1 Facilities.

ITEM	REQUIREMENT
Firing range	Selected to suit test requirements and to provide adequate protection for personnel and equipment. Firing points must be sufficiently protected and remotely located to safely contain a premature ammunition detonation or a catastrophic weapon failure.
Temperature chamber	To condition ammunition to temperatures ranging from +63° C (+145° F) to -51° C (-60° F) and humidities from 5% to 95%.
Laboratory vibration exciter	As described in TOP 1-1-050.
Drop test facility	As described in TOP/MTP 4-2-601.
Package test facility	As described in TOP 4-2-602 (App. D).
Non-destructive test facility	As described in TOP 3-2-807.
Laboratory chemical facility	To perform explosive filler exudation analysis as required.

2.2 Instrumentation for Weapon Firing Tests.

ITEM	MAXIMUM ERROR OF MEASUREMENT*
Projectile velocity measuring equipment (TOP 4-2-805)	Velocity to 3,084 m/s (10,000 fps) ±0.1%.
Weapon pressure sensing equipment (TOP 3-2-810)	Chamber pressure to 689,480 kPa (100,000 psi) ±2%.
35-mm Smear camera (TOP/MTP 4-2-816	Used to record projectile parts integrity.

*Values may be assumed to represent ±2 standard deviations; thus the stated tolerances should not be exceeded in more than 1 measurement out of 20.

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ITEM (CONT)	MAXIMUM ERROR OF MEASUREMENT (CONT)
High speed framing camera (TOP/MTP 4-2-816)	As required for recording test events.
Fall back gage	As required for recording test events.
Doppler Radar Velocimeter	As required for recording test events.
Meteorological equipment	As required for recording test events.

3. PREPARATION FOR TEST.

3.1 Test Planning.

a. As a minimum, the following tests shall be required: (the 12.2-meter and 3-meter drop tests shall be conducted prior to any other tests; and the propellant checkout and strength-of-design tests shall be conducted prior to any other firing tests).

(1) 12.2-Meter (40-Foot) Drop Test.

(2) 3-Meter (10-Foot) Drop Test (Tank Ammunition).

(3) Propellant Checkout, Assessed Service Charge.

(4) Propellant Checkout, Adjusted Charge (except for tests of the propelling charge, per se).

(5) Strength of Design (except for propelling charge).

(6) Sequential Rough Handling.

(7) Worn Tube Condition (where applicable).

(8) Vibration (Sequential).

(a) Secured Cargo.

(b) Ready-Rack (for self-propelled systems).

(9) Storage and Operation, Hot-Dry.

(10) Storage and Operation, Severe Cold.

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b. The following supplemental tests must be considered:

(1) Radio-Frequency Hazards.

(2) Fuze Arming Distance and Sensitivity.

(3) Cook-off.

(4) Intermediate Zone Firings.

(5) Fall back.

(6) Air Drop.

(7) High Humidity-Temperature.

c. Prepare a test operational checklist using the guide in Appendix A, but covering specifics for the test item and situation.

3.1.1 Test Schedules. The tests and firing conditions for different types of ammunition are summarized in Appendix B.

3.1.2 Sample Size.

a. Explosive Components.

(1) Projectiles: Minimum sample sizes for testing explosive-loaded projectiles (includes pyrotechnic, chemical and smoke rounds containing explosive dissemination devices) (depending on packaging configuration) are shown in Appendix B. The sample size is decreased if the weapon used does not show significant change in performance because of wear.

(2) Propelling Charges: Minimum sample sizes for testing propelling charges (depending on packaging configuration) are shown in Appendix B.

(3) Fuzes: Minimum sample sizes for testing fuzes which can be fired from various weapons (depending on packaging configuration) are shown in Appendix B. This sample size is valid only when a single weapon, of all those which will fire the fuze, provides the maximum conditions of setback, spin, and bore travel (in calibers).

If spin and setback are maximized in one weapon but longer bore travel will occur in another weapon, the sample size must be doubled since the fuze must be fired in both weapons to induce maximum stress levels. Similarly, if one weapon produces maximum spin, a second maximum setback and a third maximum bore travel and sample size must be tripled. The replication of sample size applies to all preconditioning cited in Appendix B (e.g., Vibration, Sequential Rough Handling, etc.). The sample size can be reduced for all weapons which do not show significant performance changes because of wear. If the fuze is designed to be used with rocket-assisted projectiles then all firing must be done with rocket-assisted projectiles with the rocket motor "on."

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b. Non-explosive projectiles. A smaller sample - The minimum sample sizes shown in Appendix B (depending on packaging configuration) - may be used for non-explosive projectiles (with spotting charge) since the potential hazards associated with the launching of HE projectiles are not present. This sample is decreased if the weapon does not show significant change in performance because of wear.

3.1.3 Extreme Temperature Limits

a. Ammunition is usually tested for safe storage, transport, handling, and weapon firing at a temperature of +63° C (+145° F), or as specified in the requirements documents or design limitations. This is the temperature equivalent to the upper extreme that ammunition will experience under operational conditions 1/ induced by the ambient air temperature and solar radiation levels of the Basic-Hot Climatic Category (A-2) and is considered to be an adequate approximation of the mean peak internal temperature attained under the storage and transit conditions of the Hot-Dry Climatic Category (A-1) of AR 70-38 2/ and STANAG 2831 3/. Ammunition is not normally tested under the Hot-Dry (A-1) operational condition.

b. Unless otherwise specified in the requirements documents or de ign limitations, the lower temperature for all safety tests shall be -51° C (-60° F) which meets the Severe Cold (C-3) requirement of AR 70-38 and STANAG 2831 for operation, storage, and transport.

3.2 Ammunition.

a. Select each ammunition/component sample from one lot. If it is not possible to select all samples from a single lot, additional lots from the same manufacturing process may be used.

b. Identify each component with a unique number to be used throughout testing. Key the numbering system to any inspection records furnished by the loading plant.

c. Make prefiring measurements of the projectile as described in TOP/MTP 4-2-800 and compare with drawing requirements.

d. Inspect all projectiles, components, and metal parts for damage, exudation, deterioration, and obvious manufacturing defects.

1/ Francis, David J., "Research Test of Solar Radiation Effects of Resultant Temperature for Certain Artillery Munitions," Aberdeen Proving Ground, MD. Report DPS-1145, Vol II, October 1964.

2/ AR 70-38, Research, Development Test, and Evaluation of Materiel for Extreme Climatic Conditions.

3/ NATO STANAG 2831, "Climatic Environmental Conditions Affecting the Design of Materiel for Use by NATO Forces Operating in a Ground Role," April 1977.

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e. Inspect projectiles for metallurgical defects using the non-destructive techniques (magnetic, particle, sonic, or fluorescent particle inspection) described in TOP 3-2-807. Radiographs are taken of explosive components and compared to appropriate standards to determine if cracks and voids are present which would contribute to a safety failure. Loading plant radiographs of acceptable quality may be used provided the munitions have not experienced significant transport and handling stresses.

f. When internal pressure gages (TOP 3-2-810) are to be used with fixed ammunition, place the gages in the cartridge case before loading. Internal gages should not be used in cartridges to be subjected to vibration or rough handling due to the potential for damage to the propellant or primer.

g. Record the following for each projectile, complete round, or component, as applicable:

(1) Lot number and manufacturer.

(2) Type, model, and caliber.

(3) Weight.

(4) Dimensions (TOP 1-2-504).

(5) Metallurgical defects (TOP 3-2-807).

(6) X-ray results for explosive-filled projectiles (usually provided by commodity command to test agency).

(7) Propellant type, composition, and web size; weight.

(8) Complete description and photographs of packaging.

(9) Date of manufacture.

3.3 Weapon.

a. For new-tube testing, select a tube for which the ammunition is standard and a tube with at least 75 percent wear life remaining. For worn tube testing select a weapon for which the ammunition is standard and a tube with a maximum of 25 percent wear life remaining or other specified wear criteria.

b. Record the nomenclature and serial number of the weapon and tube, the number of actual and EFC rounds on the tube, and the amount of wear on the tube (stargage record) as described in TOP 3-2-802.

3.4 Instrumentation.

a. Install instrumentation to measure the following in accordance with the listed references:

(1) Muzzle velocity as described in TOP 4-2-805.

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(2) Weapon chamber pressure as described in TOP 3-2-810. Use two tourmaline pressure transducers, when possible, to measure pressure-time histories at each end of the chamber for the detection of reverse pressure differentials. (Essential when propelling charge is under test.)

(3) Meteorological data per TOP/MTP 3-1-003, as required.

b. For all tests, position cameras as described in TOP 4-2-501.

4. TEST CONTROLS.

a. If required, record surface and aloft meteorological data before, during, and after each series of firing trials as prescribed in TOP/MTP 3-1-003.

b. Inspect the weapon throughout testing in accordance with the schedules of TOP 3-2-800 or more frequently when there are unexpected occurrences such as projectile breakup, cartridge case rupture, or chamber pressures 10% in excess of weapon permissible individual maximum pressure (PIMP) during a firing program.

c. Conduct all tests with the test items uniformly conditioned to the appropriate temperatures which are defined as follows: Low, -51° C (-60° F); high, +63° C (+145° F); and temperate (control), +21° C (+70° F), unless otherwise specified (para 3.1.3). Maintain temperatures to within ±1.4° C (±2.5° F). Allow sufficient temperature conditioning time to assure complete temperature stabilization. Use thermocoupled dummy rounds or charges to monitor temperature conditioning when appropriate.

d. Use a propelling charge that has been developed to meet the safety test pressure requirements. It may be necessary to add propellant to a standard charge or develop a special proof charge to achieve the desired pressure levels. Caution: If weight adjustment is required to obtain the desired pressures, coordinate with an interior ballistics specialist to insure that there will be no damage to the weapon.

e. Review radiographs and inspection results for each round, prior to firing, for abnormalities which would cause malfunctions. Some of the vibration and rough handling tests may damage the test munitions. These items will be fired if it is judged by the tester that troops in the field would have overlooked or considered the damage negligible and fired the item.

f. Continuously monitor round-by-round data to insure that critical parameters are within specified limits.

g. Indirect-fire, explosive-loaded projectiles are fired to at least 75% of the maximum range of the projectile. This is required to verify the safety of the projectile (and fuze) during long times of flight.

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h. 12.2-meter (40-foot) and 3-meter (10-foot) drop tests shall be conducted prior to any other test, if possible.

i. Propellant checkout and strength-of-design tests should be conducted prior to any other firing tests.

5. PERFORMANCE OF TESTS. Conduct the tests below as applicable, following the appropriate firing conditions of Appendix B.

5.1 12.2-Meter (40-Foot) Drop Test. An essential test for all packaged explosive items is the 12.2-meter drop which simulates an accidental drop from ship loading or unloading.

5.1.1 Method. Conduct 12.2-meter drop tests of explosive components in accordance with TOP 4-2-601. These test samples are not weapon-fired but must be safe to handle and dispose of after the drop.

5.1.2 Data Required. Collect data in accordance with TOP 4-2-601.

5.2 3-Meter (10-Foot) Drop Test. This test simulates the accidental drop of unpackaged cartridges during vehicle loading of tank ammunition. These cartridges are not weapon-fired but must be safe to handle and dispose of after the drop. This test shall be conducted after the 12.2-meter drop test and prior to any other testing. Only ammunition that is fired from tank weapons is subjected to this test.

5.2.1 Method. Subject unpackaged cartridges to 3-meter (10-foot) drop test as described in TOP 4-2-602. Inspect all components, assess any damage, and dispose of all explosive components after the drop. This ammunition is not weapon-fired.

5.2.2 Data Required. Collect data in accordance with TOP 4-2-602.

5.3 Propellant Checkout, Assessed Service Charge. This test is conducted to confirm that assessed service charge will meet the weapon and/or projectile/fuze velocity and chamber pressure requirements. Muzzle blast overpressure is also measured during this test.

5.3.1 Method.

a. Using the assessed service charge and a "new" tube, fire a ten-round group at each propellant-conditioning temperature of +21° C, +63° C, and -51° C. If the propellant checkout is part of a projectile/fuze test use the projectile/fuze that is under test. If the propellant checkout is part of a propelling charge test, if practical, use the type of projectile

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that produces the highest chamber pressure in the weapon system. For weapon-propellant combinations which have not been measured for muzzle blast overpressures, the procedures of TOP/MTP 4-2-822 shall be used to measure the muzzle blast overpressure produced by the high-temperature, weapon-firing portion of this test.

b. Make chamber pressure versus time measurements at a minimum of two locations in the chamber and determine the pressure differential as described in TOP 3-2-810. If differential pressure exceeds the levels described in Appendix D, the test should be stopped and an interior ballistic expert consulted.

c. The average peak chamber pressure at +63° C (indicated by the tourmaline gage that reads the highest) must be below the permissible individual maximum pressure (PIMP) of the weapon (or projectile) by at least three standard deviations (see Appendix C). If the charge does not meet this requirement, testing should be stopped or consideration be given to specifying a lower restrictive temperature limit for the charge.

d. The velocity at +21° C must meet system velocity requirements, and the time/pressures records at all temperatures must be examined for irregularities.

5.3.2 Data Required.

a. Date and time of firing.

b. Temperature of ammunition.

c. Muzzle velocity.

d. Weapon chamber pressure (including pressure-time and differential pressure histories).

e. Meteorological data, as required.

f. Propelling charge weight.

g. Projectile weight.

h. Weapon identification.

i. Elevation.

j. As-fired projectile weight.

5.4 Propellant Checkout, Adjusted Charges for Safety Test Weapon Firing. This test is conducted to provide data for adjustment of the charge to achieve safety-test peak pressure levels to be used when conducting high and low temperature weapon-firing phases of projectile and fuze safety tests. (When the propelling charge is the item to be tested, no charge weight adjustment is made.)

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5.4.1 Method.

a. With propelling charge uniformly conditioned at +21° C, fire five rounds using 95% of the service charge weight, and five rounds using 105% of the service charge weight from a weapon having a "new" tube. The peak chamber pressures for each of these rounds and the peak chamber pressures measured during the ten-round, +21° C, assessed service charge propellant checkout test (5.3.1a) are used to establish a second-order, least-squares prediction curve for peak chamber pressure as a function of propelling charge weight at +21° C. The prediction curves for peak chamber pressure as a function of propelling charge weight at each conditioning temperature of -51° C and +63° C are obtained from the +21° C curve by linear transformation of the temperature variable. These curves are shown as solid lines in Figure 1. These relationships are prerequisite to estimating the "adjusted" propelling charge weight for each temperature extreme which can be expected to achieve the peak chamber pressure levels which are required for the weapon-firing phases of the projectile and fuze safety tests. The maximum peak chamber pressure that is expected to occur during tactical weapon firing at each temperature extreme is estimated using the procedures of Appendix F. These pressures are used to estimate the charge weights as shown by the dotted lines in Figure 1. These estimates are then verified as follows:

(1) Fire ten rounds loaded with the high-temperature, adjusted propelling charge conditioned to +63° C using a "new" tube. The average peak chamber pressure of this group should be within ±2% of the peak safety test pressure level for +63° C established per Appendix F.

(2) Fire ten rounds loaded with the low-temperature, adjusted propelling charge conditioned to -51° C using a "new" tube. The average peak chamber pressure for this group should be within ±2% of the peak safety test pressure level for -51° C established per Appendix F.

b. If the pressures are not within the required limits, the propelling charges must be readjusted.

c. These adjusted propelling charges are used for all subsequent high and low temperature weapon-firing phases of projectile and fuze safety tests. For fixed ammunition, adjustments to standard propelling charges are made after exposure of the cartridge to the environmental test condition (vibration, shock, storage, etc.).

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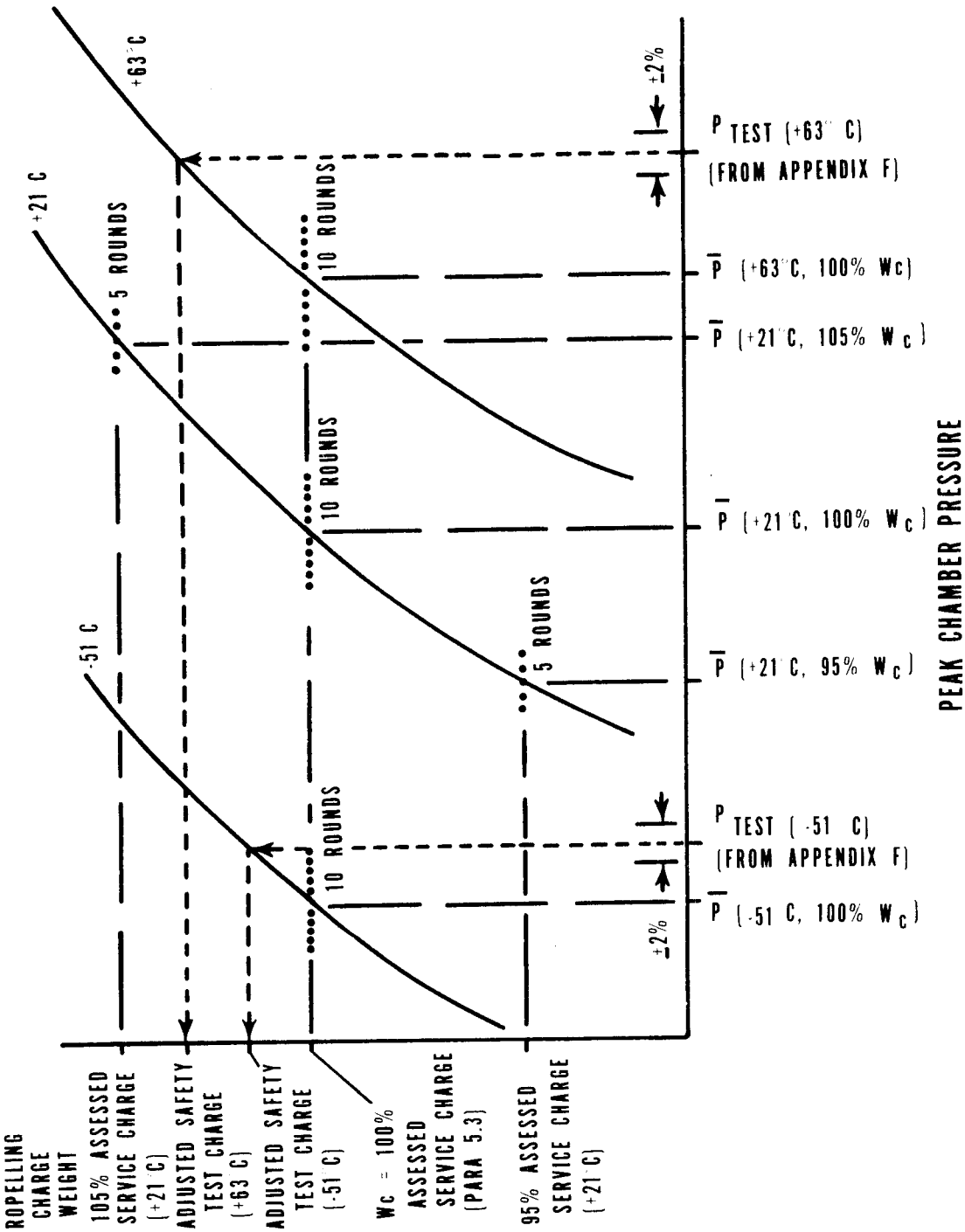


Figure 1. Estimation of Adjusted Propellant Charge Weight for Extreme Temperature Weapon-Firing Safety Tests of Projectiles and Fuzes.

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5.4.2 Data Required. Collect data as shown in paragraph 5.3.2.

5.5 Strength-of-Design Test.

5.5.1 Method. Conduct strength-of-design tests as described in TOP 4-2-501.

5.5.2 Data Required. Collect data in accordance with TOP 4-2-501 and paragraph 5.3.2.

5.6 Sequential Rough Handling Tests. Rough handling tests are conducted separately from the secured-cargo and racked-ammunition vibration test (para 5.8). Rough handling tests simulate the severe shocks, bumps, and drops that an item may receive when thrown loosely on the bed of a truck or tracked vehicle, transported unsecured over rough terrain, dropped from a tailgate, or dropped while being loaded onto a vehicle or into a weapon.

5.6.1 Method. Subject the test items to a sequence of rough handling tests in accordance with TOP 4-2-602. These tests usually consist of two packaged drops from 2.1 meters (7 feet), a loose cargo bounce test in the packaged configuration, and two unpackaged 1.5-meter (5-foot) drops with half of the test sample maintained at +63° C (+145° F) and half of the test sample maintained at -51° C (-60° F). (Separate-loading projectiles are tested bare with lifting rings installed and are not subjected to the 1.5-meter drop test. The high-temperature sample of separate-loading HE projectiles (to include submunitions-type projectiles) which are developed under the cognizance of PM for Cannon Artillery Weapon Systems will be subjected to a minimum of seven diurnal cycles of the High Temperature Storage and Low Humidity Test of TOP 4-2-820 prior to initiation of the Sequential Rough Handling Test.**) Following each phase, inspect all test items and document any damage. The rough handling tests may damage the test items. However, they will be weapon-fired if it is judged by the tester that troops in a tactical environment would have overlooked or considered the damage negligible and fired them. Uniformly condition all components suitable for weapon-firing to +63° C or -51° C corresponding to the temperature they experienced during the rough handling test, and fire in an appropriate weapon having a "new" tube. Propelling charges for weapon firing phase shall be identical with the "adjusted" charges established in paragraph 5.4.

5.6.2 Data Required. Collect data in accordance with TOP 4-2-602 and as follows:

a. Date and time of firing.

b. Temperature of ammunition.

c. Muzzle velocity.

d. Weapon chamber pressure (including pressure-versus-time and differential pressure histories unless waived).

e. Meterological data, as required.

**This is a unique requirement not applicable to other munitions and components.

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f. Photographs of projectile in flight near the muzzle.

g. Horizontal range to impact (for indirect-fire, explosive-loaded projectiles).

h. Vertical target dispersion (direct-fire projectiles).

i. Projectile weight, to include the fuze.

j. Propellant weight.

5.7 Worn Tube Condition Test. For weapons which have in previous tests showed significant changes in ballistics or launch conditions between new and worn stages, conduct additional worn tube firings as described by the firing condition tables of Appendix B.

5.7.1 Method. A sample of rounds (see Appendix B) is temperature-conditioned and fired at +63° C (+145° F) and at -51° C (-60° F). Propelling charges for this test shall be identical with the adjusted charges established in paragraph 5.4 and used for the new tube phases. For other than kinetic energy (KE) rounds, a portion of the sample projectiles at each temperature condition (see Appendix B) shall be fired inert for recovery.

For explosive-loaded projectiles, the worn tube testing should be preceded by the firing and recovery of five inert-loaded projectiles at each temperature. If recovery and examination indicate no serious body engraving or band shear, then the explosive-loaded rounds should be fired, followed by recovery of the remaining inert rounds.

Worn tube testing should be stopped under the following circumstances:

(1) Recovery of initial inert rounds shows significant body engraving or band shear.

(2) Short rounds (usually caused by excessive band shear) occur during the weapon-firing of the explosive-loaded projectiles.

It is recognized that hot worn tube chamber pressure levels are in excess of those used to establish the tube wear life (TOP 3-2-829). Thus, short rounds resulting from shearing of rotating bands does not automatically constitute a safety failure. Each incident must be examined on an individual basis and a course of action selected. Alternatives include:

(1) Lowering worn tube test chamber pressures.

(2) Changing tubes.

(3) Inserting warnings in the appropriate technical manuals, that short rounds may occur when firing at high ambient temperatures from worn tubes.

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5.7.2 Data Required. Same as paragraph 5.6.2.

5.8 Vibration Tests (Sequential).

5.8.1 Method.

a. Subject a portion of the ammunition components to be fired in the safety test (see Appendix B) to a secured cargo vibration test that simulates the logistic transportation that the rounds are likely to encounter in field use. Ammunition is vibrated in its shipping configuration. Munitions are usually boxed, though some large projectiles are mounted on pallets, and separate-loading propellants may be in cans. In any event, securely fasten the shipping module to the vibrator. Vibrate the components at +63° C or -51° C (145° F or -60° F) in two planes. Sample sizes are shown in Appendix B. Vibration schedules and durations are in TOP 1-2-601, and an explanation of laboratory vibration techniques is in TOP 1-1-050.

b. For ammunition which will be transported in the ready racks of combat vehicles, subject half of the components from each of the high-temperature and low-temperature secured cargo vibration tests to the installed equipment vibration test of TOP 1-2-601 at the corresponding temperatures of 63° C and -51° C. This test shall expose the test samples to the dynamic environment of the ready rack of the combat vehicle. Therefore, the vibration test shall be conducted with the items installed in a vibration fixture, simulating the specific ready rack in which they will be transported. Where possible, the vibration schedule to which the rounds will be exposed is developed from the vibration data obtained from tests on the same ammunition ready rack found in a specific vehicle.

c. Before vibration, X-ray all rounds or a representative sample. Following vibration, inspect the rounds, using X-rays if appropriate. Uniformly condition all acceptable components to +63° C or -51° C, corresponding to vibration test temperatures unless otherwise specified, and fire in an appropriate weapon using a "new" tube. Propelling charges for weapon firing phase of projectile and fuze tests shall be identical to the "adjusted" charges established in paragraph 5.4. The vibration tests may damage the test munitions. These items will be fired if it is judged by the tester that troops in the field would have overlooked or considered the damage negligible and fired the item.

5.8.2 Data Required. Collect prefiring test data in accordance with TOP 1-2-601 and paragraph 5.6.2.

5.9 Storage and Operation, Hot-Dry. This test simulates storage and weapon firing in desert areas.

5.9.1 Method.

a. Subject test sample as shown in Appendix B to the seven-day high-temperature/low-humidity storage cycle test described in TOP 4-2-820.

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Following storage simulation, inspect sample, using X-rays if appropriate. Adjust the charge per paragraph 5.4, uniformly condition the sample to +63° C, and fire in an appropriate weapon using a "new" tube. Propelling charges for weapon firing phase of projectile and fuze tests shall be identical to the "adjusted" charges established in paragraph 5.4.

b. For test projectiles which contain white phosphorus, a minimum of ten samples shall be oriented with the longitudinal axis of the projectiles in a horizontal plane for the seven-day storage cycle test and post-storage cooling (samples subjected to a uniform temperature of 21° C) to establish a side void. The test items should be marked to indicate the upper side during the heating and cooling cycle for reference. Following storage simulation, inspect the sample, using X-rays if appropriate, and establish projectile unbalance in accordance with TOP 4-2-801. Uniformly condition the sample to 21° C and fire it in an appropriate weapon using a "new" tube to determine whether the unbalance is sufficient to produce erratic flight. The remainder of the white phosphorus rounds will be tested as indicated in paragraph 5.9.1a. Propelling charges for weapon firing phase of projectile test shall be identical to the "adjusted" high temperature charge established in Paragraph 5.4.

5.9.2 Data Required. Collect data in accordance with TOP 4-2-820 and paragraph 5.6.2.

5.10 Storage and Operation, Severe Cold. This test simulates sub-Arctic storage and weapon firing.

5.10.1 Method. Subject test sample as shown in Appendix B to a minimum of 72 consecutive hours of storage at -51° C (-60° F). Following storage simulation, inspect sample, using X-rays if appropriate. Uniformly condition sample to -51° C and fire in an appropriate weapon using a "new" tube. Propelling charges for weapon firing phase of projectile and fuze tests shall be identical to the "adjusted" low temperature charge established in Paragraph 5.4.

5.10.2 Data Required.

a. Storage temperature record.

b. Data in accordance with paragraph 5.6.2.

5.11 Supplemental Tests. Perform the following supplemental tests as applicable in accordance with the references shown.

5.11.1 Radio-Frequency Hazards. Electronic circuitry associated with certain ammunition (e.g., electric primer or other electroexplosive device) may be susceptible to accidential initiation by stray radiation. A procedure for testing a round for hazards of initiation by electro-magnetic radiation in the r-f portion of the spectrum may be found in TOP 1-2-511.

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5.11.2 Fuzes. Fuze characteristics having a bearing on safety (i.e., arming distance, sensitivity) should be available from design test records; if not, they are determined as part of the safety evaluation in accordance with TOP/MTP 4-2-806.

5.11.3 Cook-Off. The tendency of propelling charges to cook-off should be examined. This is especially true for new propellants. Time to cook-off versus tube temperature should be established through the use of a modified tube which can be heated to uniform temperature with a heating coil.

5.11.4 Intermediate Zone Firings. In rare instances, a specific projectile may be considered more susceptible to safety failures at low and intermediate zones than it is at the top zone even though traditional interior ballistic parameters of velocity, pressure, acceleration, and spin are less. This possibility may be considered in the design review portion of the tests planning phase. If past experience or engineering judgement suggests this possibility, appropriate additional tests will be designed and conducted.

5.11.5 Fallback. Fallback of separate loading artillery projectiles onto the propelling charge can cause a catastrophic failure when the round is fired. Fallback is generally associated with improper ramming; however, there have been several instances where fallback was the result of incompatibility between the projectile rotating band and the forcing cone of the tube. A procedure for determining projectile retention characteristics is contained in TOP 4-2-802. This procedure also describes the use of a strain gage to assure that fallback has not occurred. This fallback gage should be used during all safety testing of separate loading ammunition to assure that fallback has not occurred thus eliminating fallback as a factor if a malfunction occurs.

5.11.6 Airdrop. If the items are designed for airdrop, then airdrop safety testing may be required. Generally the forces experienced during rough handling exceed the forces expected during high velocity impact of air dropping munitions. However, the rigging/packaging of the items may allow airdrop impact forces to exceed rough handling forces. These areas should be reviewed, and high velocity airdrop test or low altitude extraction tests planned if necessary. MTP 4-2-509 describes airdrop testing.

5.11.7 High Humidity-Temperature. In some cases, high humidity environment can be an important aspect of the safety evaluation. One example is combustible cartridge cases where moisture can alter burning characteristics, create smoldering residue and make it unsafe to load subsequent rounds. Conduct test as described in TOP 4-2-802.

6. DATA REDUCTION AND PRESENTATION.

a. Assemble and tabulate all results and safety information generated during the preliminary safety tests conducted in accordance with this TOP

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and the performance tests conducted as described in the test plan. Assign the proper hazard level and probability to each hazard in accordance with MIL-STD-882A. Report hazard level and classification (deficiency, shortcoming, etc.) in accordance with DARCOM R 700-38 (paragraph 4f) and TECOM Supplement 1 4/ and DSAR 8220.1 5/. Report the conditions of use under which each hazard was observed and describe any features that require further investigation, including any hazards that could occur or increase as a result of storage. Describe (narratively) all hazards identified and recommended actions required to eliminate or avoid each potential hazard.

b. Determine the safety of ammunition components from the following considerations:

(1) Launch safety - A projectile or fuze will normally be considered launch-safe if no prematures occur either in-bore or within the arming limits of the fuze. In-bore premature is defined as any functioning of the projectile while it is inside the weapon.

(2) Flight safety - A projectile or fuze will be considered flight-safe if no early function occurs after fuze arming or before the end of the flight. In addition, there shall be no distortion or separation of metal parts, or evidence of erratic flight or short rounds. Rounds which impact more than four probable errors in range or deflection from the center of impact shall be reviewed to determine the source of this atypical behavior. Additional weapon firing trials may be required to verify the analyses.

(3) Compatibility with design limits of weapon - The chamber pressure characteristics of a complete round at high temperature will not exceed those of the design criteria for the weapon, and no significant differential pressures will develop. An upper safe operating temperature for the round will be specified in the safety release recommendation. A procedure for determining this upper temperature limit is described in Appendix C.

(4) Transportation, rough handling, storage safety - Ammunition components under test will be considered safe for transportation, handling, and storage under the conditions tested, provided that the test conditions did not induce any safety problems.

c. Examine the safety of non-explosive munitions for each of the factors prescribed for explosive rounds. In the area of launch safety, the significance of in-bore failure of a non-explosive round may be different from that of an HE round, but must also be considered in terms of the likelihood that in-bore failure may produce a weapon failure or create a condition that will endanger personnel. The weapon failure can

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4/ DARCOM R 700-38, Test and Evaluation - Incidents Disclosed During Materiel Testing.

5/ DSAR 8220.1, Explosive Hazard Classification Procedure.

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be catastrophic as in the case of a propelling charge that develops sufficient pressure levels to induce breech failure.

d. Evaluation of pressure-time and differential pressure records will be analyzed in accordance with Appendix D to determine propelling charge safety.

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APPENDIX A
GUIDE FOR TEST OPERATIONS CHECKLIST

ITEM	YES	NO	NA
1. All operating personnel briefed on test requirements, special procedures, hazards and any unusual aspects of test.			
2. Test items are inspected, available, and numbered.			
3. All instrumentation calibrated, properly installed, and operational.			
4. Facilities ready (itemized for convenience).			
5. Required preliminary data recorded.			
6. Safety SOP's prepared and posted.			
7. Other.			

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APPENDIX B
TEST PHASES FOR SAFETY TESTING OF ARTILLERY, MORTAR, AND RECOILLESS RIFLE AMMUNITION

Test Phase Sequence (Independent Samples)	Separate-Loading Ammunition (Number of Rounds)									
	Explosive Projectile Test			Fuzes (8/Box)			Propelling Charge Test			
	Hot	21° C	Cold	Hot	21° C	Cold	Hot	21° C	Cold	
1. 12.2-Meter (40-Foot) Drop Test TOP 4-2-601	5*		5*	6**		6**	6**			6**
2. Not Applicable										
3. Propellant Checkout Assessed Service Charge (Inert Fuzes and Projectiles)	10	10	10	10	10	10	10	10		10
4. Propellant Checkout Adjusted Charge (Inert Fuzes and Projectiles)	10	10	10	10	10	10				
5. Strength of Design (Inert Fuzes and Projectiles) TOP 4-2-501		10			10					
6. Sequential Rough Handling TOP 4-2-602	30		30	48		48	35			35
7. Worn Tube Condition										
High Explosive	50		20	48		24				
Inert Recovery	10		10							
8. Vibration (Sequential)										
Secured Cargo (Packaged)	50		25	48		24	50			50
Ready-Rack (Bare) (Half of Secured Cargo Samples)	25		12				25			25
9. Storage, Hot-Dry TOP 4-2-820	50			48			50			
10. Storage, Cold			25			24				50
11. Supplemental Tests	As Required			As Required			As Required			

*Bare Projectiles
**Packages

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Test Phase Sequence (Independent Samples)	Packaged (Fixed) Ammunition (Number of Rounds)								
	Explosive Projectile and Fuze Test			Non-explosive Projectile Test			Propelling Charge Test		
	Hot	21° C	Cold	Hot	21° C	Cold	Hot	21° C	Cold
1. 12.2 Meter (40-Foot) Drop Test Packages) TOP 4-2-601	6		6				6		6
2. 3-Meter (10-Foot) Drop Test (Tank Ammunition Only) TOP 4-2-602	5		5	5		5	5		5
3. Propellant Checkout Assessed Service Charge (Inert Fuzes and Projectiles)	10	10	10	10	10	10	10	10	10
4. Propellant Checkout Adjusted Charge (Inert Fuzes and Projectiles)	10	10	10	10	10	10			
5. Strength of Design (Inert Fuzes and Projectiles) TOP 4-2-501		10			10				
6. Sequential Rough Handling (Sample Size depends on package configuration) TOP 4-2-602	35/80		35/80	35/80		35/80	35/80		35/80
7. Worn Tube Condition									
High Explosive	50		20						
Inert Recovery	10		10	10*		10*			
8. Vibration (Sequential)									
Secured Cargo (Packaged)	50		25	20		10	50		50
Ready-Rack (Bare) (Half of Secured Cargo Samples)	25		12	10		5	25		25
9. Storage, Hot-Dry TOP 4-2-820	50			10			50		
10. Storage, Cold			25			10			50
11. Supplemental Tests	As Required			As Required			As Required		

*Other than KE projectiles.

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APPENDIX C
METHOD FOR ESTABLISHING MAXIMUM SAFE TEMPERATURE LIMITS FOR ARTILLERY PROPELLANT SYSTEMS

Before a weapon system is released for operational testing, the extreme-temperature limitations applicable to the system in combination with its particular ammunition must be established. In determining the temperature increase that can be allowed without exceeding weapon or projectile pressure limits, it is necessary to:

a. Determine the pressure level of the charge for service velocity at 21°C (+70°F), This should be based on the highest zoned weight of the heaviest projectile.

b. Establish the pressure variability that may be expected from production components associated with the system (lot-to-lot, gun-to-gun, etc.).

c. Determine the relationship between pressure and temperature for the system.

The expected pressure for the service velocity can be obtained from propellant assessment results obtained during development testing. While new tube data are most desirable to eliminate the effect of tube wear on velocity and pressure, it is possible to obtain a usable velocity-pressure relationship from other than new tube firings. Regardless of the velocity-pressure source, service pressure must always be lower than the upper pressure limit for the propellant proof by a margin consistent with propellant production and assessment capabilities. A margin of approximately 10,342 to 13,790 kPa (1500 to 2000 psi) will satisfy most systems.

The second area of consideration, following the service pressure level determination, is to define other factors that affect pressure and to obtain a quantitative evaluation of their effect. Pressure variability is attributable primarily to round-to-round differences and to differences lot-to-lot, tube-to-tube, and occasion-to-occasion. While an estimate of round-to-round dispersion can be obtained from development tests, it is necessary to use data from the firing of older weapon systems in estimating the effects of other variables. Propellant calibration tests present a reasonable quantity of data under controlled conditions. Table C-1 shows pressure dispersions in terms of standard deviation (σ) associated with round, tube, and occasion based on calibration firings for many existing systems.[6/]

6/ Anderson, H. B., "A Study to Summarize Pressure Characteristics for Artillery Weapons Systems," Aberdeen Proving Ground, Md., Report DPS-1648, January 1966.

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TABLE C-1. PRESSURE DISPERSIONS

Weapon	Press., Std Dev psi			
	Round	Lot	Tube and Occasion	Total
High performance tank weapon	645	452	471	934
Medium to long range field artillery	501	452	445	836
Mortars	272	272	146	414
Recoilless rifles	184	130	48	229

Propellant lot-to-lot pressure variations for production lots tested against calibration propellant lots are tabulated in reference 5.

These dispersion estimates are combined as follows:

$$\sigma_T = \sqrt{(\sigma_r)^2 + (\sigma_a)^2 + (\sigma_t)^2}$$

where

σ_T = total pressure variation for new charge.

σ_r = round-to-round pressure variation (obtained from firings of new charge.)

σ_a = ammunition lot-to-lot pressure variation (obtained from old charge data.)

σ_t = combined tube and occasion pressure variation (obtained from old charge data.)

From these data, $3\sigma_T$ is interpreted as providing an estimate of the variability of future production lots of the new charge and established 3σ limits around the expected service pressure to include practically all rounds with any propellant lot fired in any tube or on any occasion.

The third requirement in determining the temperature for the charge is an expression showing the pressure-temperature relationship of the charge. For details for establishing the temperature coefficient see TOP/MTP 4-2-608.

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With the preceding information the required temperature limit can be determined as follows:

Let P_s = expected service pressure, kPa (psi).

$3\sigma_T$ = total pressure variation, kPa (psi).

K = pressure-temperature coefficient, kPa/^{o}C (psi/^{o}F).

W PIMP = Weapon Permissable Individual Pressure limit.

Then $P_M = P_s + 3\sigma_T$ = expected maximum permissible individual round pressure at 21° C (70° F).

P_T = W PIMP - P_M = pressure margin available for temperature increase.

$\frac{P_T}{K} = \Delta T$ = allowable temperature increase.

21°C (70°F) + ΔT = temperature limit

This temperature limit is assumed to be applicable to new tubes where pressures are expected to be highest.

When improved information on the above factors becomes available from firings with a specific charge, a revised temperature limit should be computed.

The procedure for determining the maximum safe temperature for projectiles having lower design pressure than the weapon is essentially the same. The projectile design pressure, however, rather than the weapon design pressure, is the constraint.

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APPENDIX D
PROCEDURES FOR EVALUATION OF SAFETY OF PROPELLING CHARGES

Evaluation of pressure-time records provides the basis for determination of charge safety. The flow chart at figure D-1 depicts the events relating to the determination of charge safety. 7/

STEP 1: Velocity and pressure data are analyzed for evidence of erratic performance, i.e., long ignition delays (hangfires), high negative differential pressures, inflections in the pressure-time trace occurring prior to peak pressure, high peak chamber pressures or high muzzle velocity standard deviations. If none of these indications are present, the charge may be presumed to be safe and a safety release recommendation may be issued.

The following guidelines may be used to determine if pressure-time anomalies exist; they are not intended to be universal and may not be appropriate for some propelling charges. If the mean and standard deviation of the initial negative differential pressure ($-\Delta P_i$) are less than or equal to 1,000 psi (6,900 kPa) and the individual maximum $-\Delta P_i$ does not exceed 5,000 psi (35,000 kPa) the charge may be considered safe and a safety release recommendation issued. These limits are applicable to propelling charges with operating pressures in the range of 30 to 50 ksi (206,800 to 349,700 kPa) which do not exhibit low temperature sensitivity. 8/ At higher operating pressures these limits may be too loose. Performance data generated during engineering development of the charge will be useful in determining the applicability of these limits to a particular propelling charge.

A propelling charge which exceeds the general guidelines is not necessarily unsafe, but it will be necessary to generate specific guidelines for the particular charge as described in steps 2 through 5.

STEP 2: Additional firings are conducted with altered or misassembled propelling charges to deliberately induce high differential pressures in order to determine their effect on the peak pressure level (sensitivity). The sensitivity may be related to one or more of the following factors.

7/ Herud, Craig, "Methodology Investigation, Techniques for Testing and Evaluating the Safety and Performance Characteristics of Propelling Charges," Aberdeen Proving Ground, MD, Report No. APG-MT-5136, February 1979.

8/ Clarke, E. V., May, I. E., Kelso, J. R., "Effects of Pressure Wave Dynamics on the Ballistic Performance of Guns," US Army Ballistic Research Laboratory, Aberdeen Proving Ground, MD, Interim Memorandum Report No. 311, November 1974.

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a. Operating pressure of the charge.

b. Temperature of the propellant.

c. Type of igniter (center core or base ignited).

d. Propellant type and web size.

e. Loading density.

f. Charge standoff from the primer (for bagged charges).

g. Primer output.

STEP 3: Data from firings with induced defects are plotted along with data from normal charge firings (peak pressure versus differential pressure) and a sensitivity relationship is established. The sensitivity data are examined to determine if a particular propelling charge exhibits extreme sensitivity to differential pressure i.e., a small increase in $-\Delta P_i$ produces a large increase in peak pressure. If the charge exhibits very little or no sensitivity, a safety release may be issued. If the charge is sensitive to differential pressures, the distribution of both the peak chamber pressures and the differential pressures must be examined in detail.

STEP 4: Additional firings are conducted if necessary and test data on the propelling charge are examined. A sample distribution believed to be representative of the expected performance of fielded charges is determined for both the differential pressure and the peak chamber pressure. The probability of a charge producing either a peak pressure which exceeds the weapon or projectile pressure limits or a differential pressure which will lead to a pressure which will damage the weapon (determined from sensitivity relationship) or projectile is computed.

STEP 5. If this probability is sufficiently low for either temperature extreme, generally less than 1 in 10^6 for both conditions, the charge can be considered safe and a safety release recommendation issued. If the probability of either or both of the events is significantly greater than 1 in 10^6 one of two courses of action is required: The charge must be redesigned or additional tests must be conducted to determine the sensitivity of the propelling charge to temperature. This second alternative is available only if the operational temperature limits of the charge can be changed. If they cannot, the charge must be redesigned.

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STEP 6: Additional firings are conducted at temperature extremes other than the originally specified limits. One or both extremes may be adjusted as necessary. These firings will usually include some altered charge firings to permit the determination of pressure-differential pressure sensitivity at the new temperature limits.

STEP 7: The data generated in step 7 are examined. If insufficient improvement is observed, or if significant improvement is observed at some unacceptable temperature limit, the propelling charge must be redesigned or terminated. If significant improvement is noted at some acceptable temperature limit, the propelling charge safety test is repeated at these limits and the data are evaluated as described in step 1.

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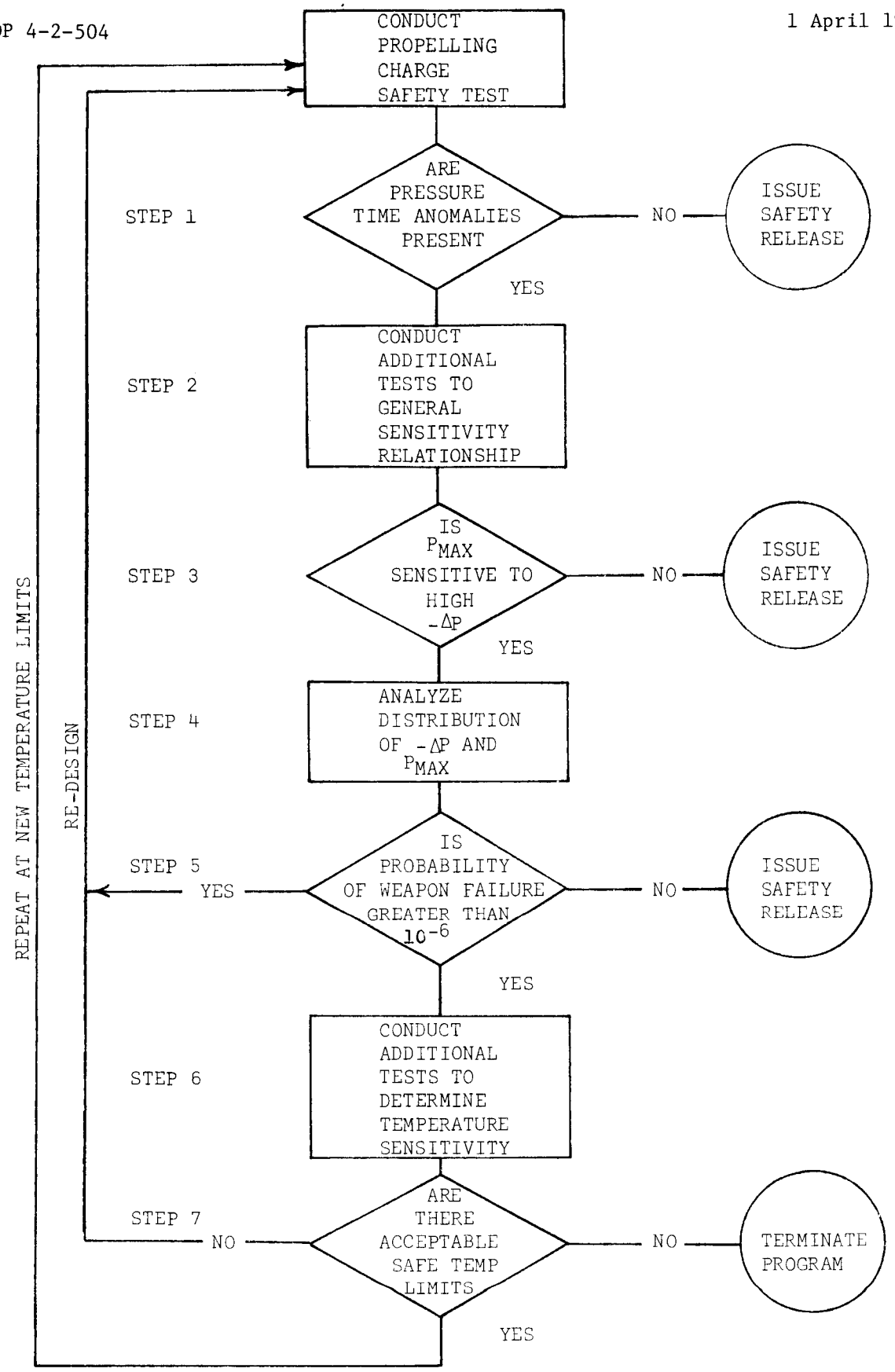


Figure D-1. Flow Chart for Propelling Charge Safety Evaluation.

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APPENDIX E
BACKGROUND

1. Introduction.

a. Assessment of munitions safety is a continuous process. Initially, it is necessary to establish that the design is inherently sound and to develop test data to show that it is safe for use during operational testing. If the item is type classified and production is initiated it is necessary to establish that changes to simplify production and the production process itself have not compromised the item's safety. This is extremely important as TECOM is responsible for issuing a statement of suitability of the item for issue (DARCOM Reg 700-34, 15 Jun 78). Confirmation of safety is a major consideration in development of TECOM's position. Finally, as product improvements are planned for incorporation into the production process it must be verified that the improved item is as safe as its predecessor.

b. The safety test is not designed to satisfy military characteristics. Military characteristics generally prescribe "premature" rates so low that no reasonable test can validate them. The safety test is designed to confirm the hypothesis of a low premature rate which is supported by related data, engineering judgment, and a moderate amount of new firing data.

c. The safety of munitions as examined in this procedure is oriented toward an identification of the hazards associated with handling, transportation, launching and flight to target. Each component of the complete round is examined, and its possible contribution to launch and flight hazards is determined. The end product is a recommended Safety Release for the munition as the user will receive it.

2. Test Design Criteria. Criteria for testing the test item must be based primarily on engineering judgment and use of the following test design considerations:

a. Statistical Considerations - Designing a test to assure high confidence levels that no more than a specified number of failures will occur is a statistical problem. When very low failure rate requirements are coupled with high confidence level requirements, very large sample sizes are needed (MTP 3-1-002). For example, the least number of samples required to provide a 95 percent confidence limit of no more than one failure per ten thousand rounds (an unfavorably high rate for prematures) would be thirty thousand rounds. Since such tests are not economically practical, other approaches besides statistical are advisable.

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b. Design Review - If the basis for expecting an acceptably low failure rate can be obtained from (1) previous related tests or (2) design considerations, then a relatively limited test program may be conducted to confirm this expectation. These considerations establish the first requirement imposed on the test director: to perform a thorough review of all data related to the item under test. If this review shows that conformity to proven design and performance in earlier (engineering design or component development) tests of this or similar items provides a basis for expecting satisfactory performance, then the procedures of this top are recommended. If not, the test plan must be expanded to provide the necessary assurance. In either event, reference to the documents reviewed and to any novel design features must be given in the test plan and in the test report.

c. Safety Statement - DARCOM Regulation 385-12 requires the submission of a safety statement from the developer prior to the commencement of the DT I, DT II or DT III. Review of the safety statement is mandatory in connection with the design review prescribed in b, above.

Additionally, if new or modified explosives are incorporated into the munition the developer's safety statement must contain the results of testing required by DA Technical Bulletin TB-700-2, Explosives Hazards Classification Procedures.

d. Inspection for Incipient Failure - For each round that produced an actual failure there are usually many that produce "near failures;" consequently, another means of compensating for sample size limitations is to obtain detailed inspection information before, during, and after testing when such data can provide indications of incipient failure. Such information can be obtained by projectile measurement before and after firing, by magnetic particle inspection for cracks, by in-flight photographs for detection of breakup, and by time-chamber pressure records.

e. Increased Severity Testing - A technique used to arrive at economical sample sizes is that of testing under conditions of increased severity. The likelihood of malfunctions arising from shock or structural failure is increased by testing under conditions representative of, or slightly exceeding, the upper extremes encountered in the field. In fairness to the designer, conditions selected must not be so extreme as to exceed limits imposed by the designer or user.

f. Generally, because of the increased severity associated with the safety test, satisfactory performance of the test item is not required. However, if the climatic performance requirements of the test item are

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within the limits used for the hot cycle and cold soak phases, performance data can be gathered and evaluated. Additionally, poor performance after transportation, vibration or rough handling may indicate a need for further investigative testing.

g. The safety test may reveal some limitations of a hazardous nature such as environmental. (An item may be "safe" only within certain temperature extremes, shock and vibration limits, and so on). In such cases the item may be released with special restrictions.

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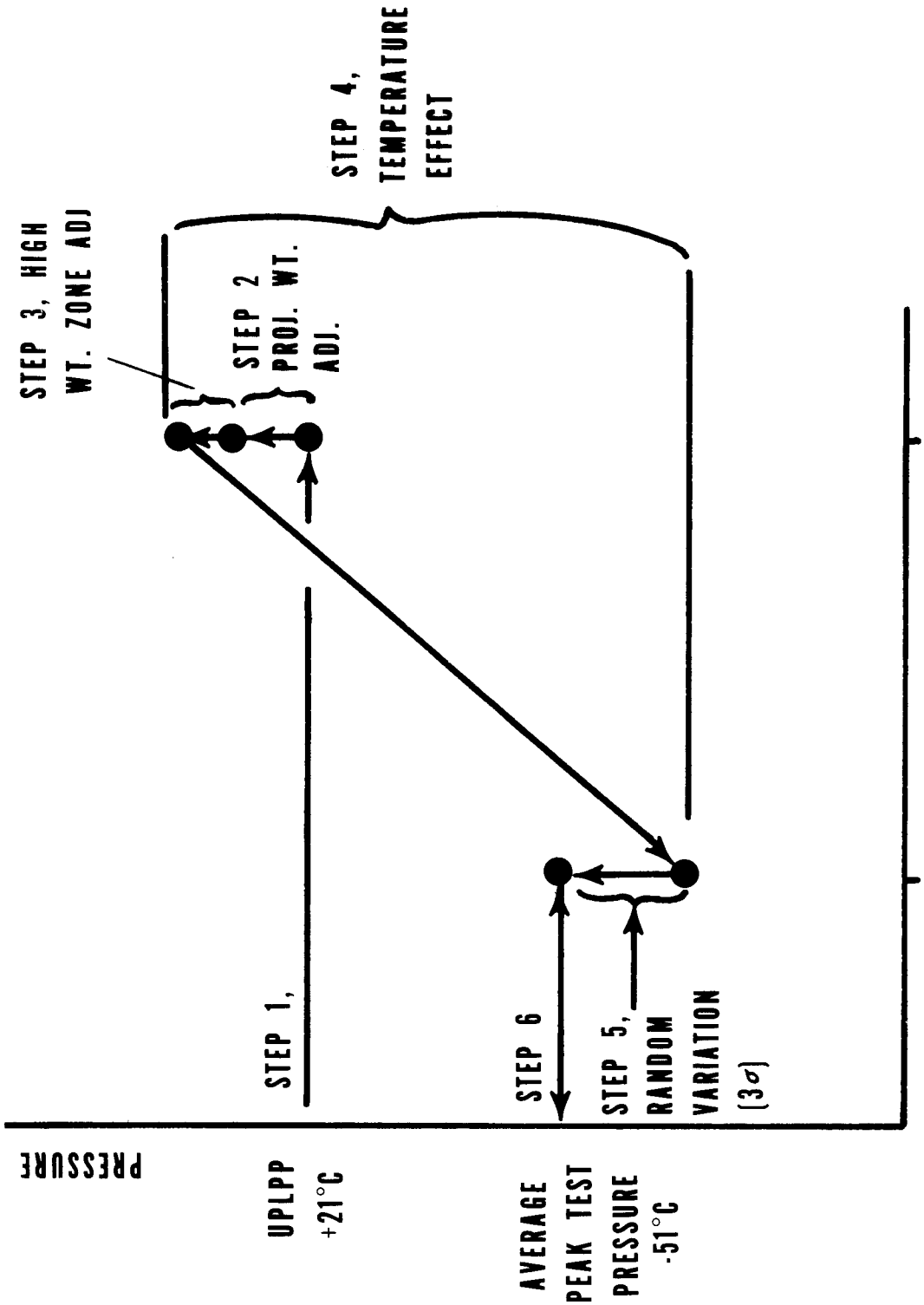


Figure F-2. Determination of Average Peak Test Pressure Level for Propellant at -51° C.

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APPENDIX F
ESTABLISHMENT OF PRESSURE LEVELS FOR SAFETY TEST FIRING OF PROJECTILES AND FUZES

1. INTRODUCTION. It has been proven that extreme temperature coupled with high launch pressure will cause failures if the test item design is deficient. Thus, the safety test must be structured to fire a limited number of items at extremes of temperature and pressure to gain confidence that the design is sound.

Since system operating temperatures are governed by AR 70-38 (or design limitations), the establishment of test pressure levels is one of the most critical aspects of the safety test. System pressure levels depend on or are limited by a number of factors including weapon design limits, projectile (or fuze) design limits or propelling charge assessment limits.

2. SAFETY-TEST PRESSURE BASED ON PIMP. If the military specification for the system's propelling charge is available, it should be reviewed to determine the allowable Permissible Individual Maximum Pressure (PIMP). This value is selected as the average peak pressure to be obtained when using extra propellant during the 63° C (145° F) safety test weapon firing. Should the projectile type used in the safety test be different from the projectile type stipulated in the propellant specification, the PIMP must be modified to reflect any difference in weight, chamber volume, rotating band, obturating band or any other difference that might exist. Any such difference will change the adjusted charge weight. The PIMP can be used only for high temperature tests. For low temperature tests, see Paragraph 3 below

3. SAFETY-TEST PRESSURE BASED ON UPLPP. If the propellant PIMP is not identified, then the upper pressure limit for propellant proof (UPLPP) can be used to determine the required pressure level. The value for the upper pressure limit for propellant proof at +21° C (70° F) should be identified in the propellant specification. If the propellant specification is not available, it may be possible to obtain this value from the propellant developer. Since the upper pressure limit for propellant proof is specified at +21° C, it must be adjusted for various factors.

a. High Temperature Tests. The +63° C pressure level for safety testing is based on the highest pressure that is expected to occur during tactical weapon firing at this temperature. If the upper pressure limit for propellant proof is established, it can be used to estimate the average peak pressure to be obtained during the +63° C firing. The following method is illustrated in Figure F-1.

Step 1: Idneitify the upper pressure limit for propellant proof at +21° C.

Step 2: Adjust this level to reflect any weight difference between the type of projectile under test and the type of projectile stipulated in the propellant specification. (This adjustment may be up or down.)

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Step 3: Adjust the resulting level upward to account for the highest zoned weight of the type of projectile under test.

Step 4: Adjust the resulting level upward to account for the increase in pressure resulting from increasing the propellant temperature from 21° C to 63° C. This adjustemnt is based on a temperature coefficient expressed as pressure/degree of temperature. The temperature coefficient may be available from other tests. If not, it has to be established as part of the propellant checkout phase of the safety test.

Step 5: Adjust the resulting pressure level upward to account for other factors that affect pressure. These factors or random variations are represented by 3 standard deviations of the combined variations due to round-to-round, tube-to-tube, and occasion-to-occasion. The total variation is determined as follows:

$$3\sigma_{Total} = 3\sqrt{\sigma^2_{Tube} + \sigma^2_{Occas} + \sigma^2_{Rd}}$$

Additional information on determination of the variations is contained in Appendix C.

Step 6: The resulting estimate is established as the average peak pressure level to be obtained during the +63° C safety test weapon firing.

If no propellant specification has been prepared, then the proof pressure cited in the weapon specification may be used as the average pressure to be achieved during the +63° C firing. This approach is acceptable only of the projectile or fuze to be tested is designed for use at the weapon limit.

b. Low Temperature Tests. The -51° C (-60° F) pressure level for safety testing is based on the highest pressure that can be obtained at this temperature. If the upper pressure limit for propellant proof is established, it can be used to estimate the average peak pressure to be obtained during the -51° C firing. The following method is illustrated in Figure F-2.

Step 1: Identify the upper pressure limit for propellant proof at +21° C.

Step 2: Adjust this level to reflect any weight difference between the type of projectile under test and the type of projectile stipulated in the propellant specification. (This adjustment may be up or down.)

Step 3: Adjust the resulting level upward to account for the highest zone weight of the type of projectile under test.

Step 4: Adjust the resulting level downward to account for the decrease in pressure resulting from decrease in temperature from 21° C to -51° C.

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Step 5. Adjust the resulting pressure level upward to account for random variations (3σ) as described previously and illustrated in Appendix C.

Step 6. The resulting estimate is established as the average peak pressure level to be obtained during the -51° C safety test weapon firing.

4. PROPELLING CHARGES. The required pressure levels can be achieved in one of two ways. It may be possible to add propellant to a standard (or developmental) charge so that when it is temperature conditioned (to +63° C or -51° C) and fired, it will produce the required pressure. If this is not possible because of volume constraints or other factors, a special or "proof charge" must be developed. Caution must be exercised in the development of a proof charge as it must operate with the pressure-time constraints of the system. A proof charge, which produces the required peak pressure, is unsatisfactory if the corresponding muzzle velocity does not result in maximum spin levels. An alternative method to obtain maximum spin level is acceptable. Additional guidance can be found in TOP 4-2-700.

5. PRESSURE LEVELS BELOW WEAPON DESIGN. Pressure levels for testing projectiles, fuzes or cartridges designed to a lower pressure limit than the weapon should be based on the designer's limits. The approach of establishing a +21° C level and then adjusting about that level as decscribed above may be used in determining these pressure levels.

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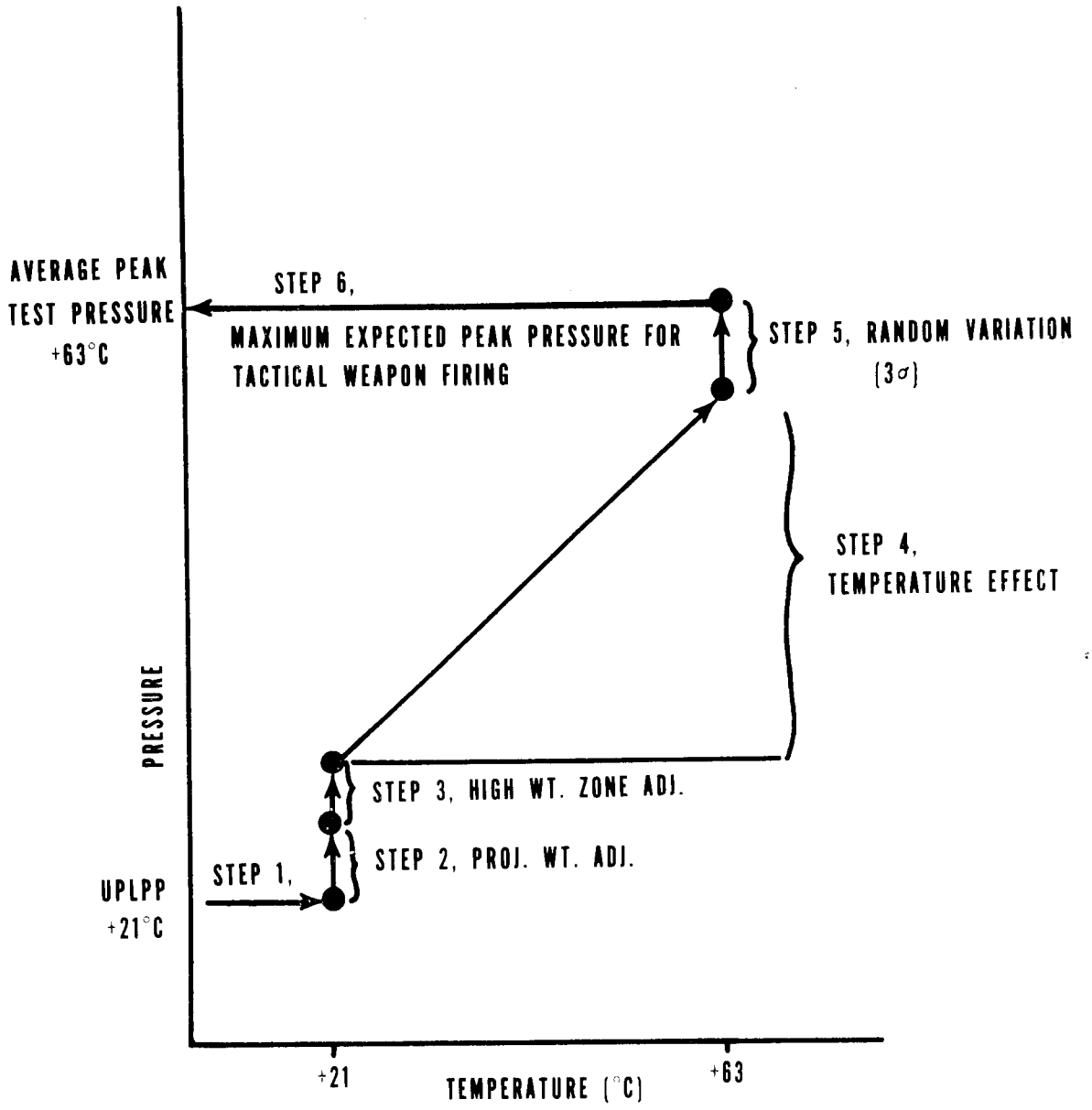


Figure F-1. Determination of Average Peak Test Pressure Level for Propellant at +63° C.

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APPENDIX G
PRESSURE DEFINITIONS 9/

1. BALLISTIC DESIGN PRESSURE CURVE is the highest average pressure for any acceptable lot of service propelling charges where conditions of propellant temperature, type and weight of projectile, and state of the cannon is specified. It is the basis of the CANNON FATIGUE DESIGN PRESSURE CURVE.

2. CANNON FATIGUE DESIGN PRESSURE CURVE represents the pressure at each point in the tube for a specified fatigue life. It is based on ammunition type, test conditions and desired reliability.

3. ELASTIC STRENGTH PRESSURE CURVE represents that pressure at each point in the tube, at which one round will cause permanent bore enlargement with material of minimum yield strength and quality.

4. PERMISSIBLE INDIVIDUAL MAXIMUM PRESSURE CURVE is the pressure at each point in a cannon tube which according to probability laws should not be exceeded by more than 13 rounds of ammunition in 10,000 under the most extreme service conditions. It represents a maximum value covering occasion-to-occasion, tube-to-tube, lot-to-lot, and round-to-round variations. It is based on the maximum allowable ammunition temperature, heaviest allowable weight for the specified projectile, and state of the cannon.

5. CANNON PROOF PRESSURE CURVE is the pressure at each point in the cannon tube utilized in cannon and tube proofing.

6. UPPER PRESSURE LIMIT FOR PROPELLANT PROOF (Formerly called Rated Maximum Pressure) is that value of the pressure which is specified in the propellant specification as the upper limit of average pressure at 70°F (21°C), which may be developed by an acceptable propellant, in the form of propelling charges, which will impart the specified muzzle velocity to the specified projectile at its standard zone weight, under stated cannon conditions.

7. LOWER PRESSURE LIMIT FOR PROPELLANT PROOF is that value of pressure which is specified in the propellant specifications as the lower limit of pressure at 70°F (21°C), which may be developed in an acceptable propellant, in the form of propelling charges, which will impart the specified muzzle velocity to the specified projectile at its standard zone weight, under stated cannon conditions.

9/ Anderson, H.B., "Pressure Definitions for Design and Test of Guns and Ammunition Systems" Aberdeen Proving Ground, Md., Report No. 4052, 20 March 1972.

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8. PROJECTILE DESIGN PRESSURE is the value of the pressure on the base of the projectile which is used for the mechanical design of the projectile.

9. PROJECTILE PROOF PRESSURE CURVE is the pressure at each point in the cannon utilized in projectile proofing.

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